

The Hong Kong Polytechnic University

Subject Description Form

Please read the notes at the end of the table carefully before completing the form.

Subject Code	SFT543
Subject Title	Introduction of Intelligent Wearable Systems
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	N/A
Objectives	Intelligent wearable technology is regarded as an emerging disruptive technology to traditional apparel or electronics. In addition to their traditional features, intelligent wearable systems have high levels of intelligence, with many new functions enhancing our everyday life. They can effectively see, feel, listen, talk, learn new things, memorize information, repair themselves, change their colour, shape or pattern, communicate with others via electronic media, mobile telephones and the Internet, lift weight and even provide their own power. This subject aims to introduce and help students understand two approaches to construct an intelligent wearable system. The first is to use fibre- or flexible film-based devices by building nanostructures on the surface or inside soft materials, like textiles or leather or plastics, which lead to a high level of ubiquitous-ness. The second route is to embed microelectronic devices in wearable products or their building elements like yarns or fabrics, which has the advantage of the utilization of the existing strength of several industries, apparel, footwear, jewellery and electronics. From progress on the course, students will begin to obtain a full understanding of the concepts, systems architectures, constituent components of rigid microelectronic or flexible and soft devices, software for control and communications, as well as applications in healthcare, sports, education, protection and others. In addition, the students will gain a glimpse of several important issues, including performance evaluation, policy, ethics, safety, standards, and regulations, as well as a market entrance requirement.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: (a) understand the architecture of intelligent wearable systems and constituent components as well as their function requirements

	(b) understand the characteristics of various innovations and advances in textiles and fashion, materials science, microelectronics etc. related to intelligent wearable technology, that define eventual potential applications across a multitude of fields (c) understand a range of principles in science and engineering in order to make well-founded decisions as a part of a design process for intelligent wearable systems in new applications (d) understand the use of appropriate candidate technologies in the planning and utilisation steps towards a desired output, in accordance with a given set of target brief specifications (e) develop critical and creative “out of the box” thinking skills in analysing, explaining and ideally matching innovations in intelligent wearable systems to specific textile technology applications, based on insights and knowledge gained from lectures, readings and own research in the subject
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	This subject will be delivered via interactive lectures, individual and group projects conducted 1. Introduction to Intelligent wearable systems + Course overview + Project assignment formats & details 2. Definitions and classifications of intelligent wearable systems¹⁻³ Wearable computers, wearable electronics, smart clothes. virtual reality, augmented reality, implantable, skin mountable, apparel and bodywear. 3. Architecture of intelligent wearable systems²⁻⁴ Hardware components and functions; software components and functions; connections and communications; body area sensor network 4. Rigid hardware components^{4,5} Sensor, display, antenna, connector, memory, central processor unit, communication chips, actuator, power supply 5. Polymer film-based and fibre-based hardware components⁵⁻⁸ Sensor, display, antenna, connector, memory, actuator, power supply 6. Performance evaluation and reliability² Test methods and standards 7. Case studies of intelligent wearable systems⁹⁻¹³ Healthcare; Education and training; sports, industrial uniforms and others, development trends 8. Related issues^{1,2} Policy, ethics, market, safety and regulations
Teaching/ Learning Methodology <i>(Note 3)</i>	• Lectures will use various visual and written references to facilitate student’s understanding of the subject • Summaries of each lecture will be distributed to students on a weekly basis

	• In order to engage student-centred learning, there is a requirement to prepare pre-class material that is ready for subsequent discussion (~3 mins) related to topics to be covered during the lecture – random students/groups will be chosen to present to the whole class. • Individual and group projects are required to reinforce understanding of the lecture contents and to promote the development of self-learning ability. • Substantial mandatory reading assignments will be given weekly, relating to the lecture topic of contemporary interest																																															
Assessment Methods in Alignment with Intended Learning Outcomes <i>(Note 4)</i>	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Group Project Assignment</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Individual Project Assignment</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Pre-class preparation, attendance & participation</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Exam</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: GROUP PROJECT ASSIGNMENT & IN-CLASS PRESENTATION: 20% In small teams, students will build a technical poster regarding a chosen topic in intelligent wearable systems. Subsequently, students will also provide a short oral presentation overview (including a Q&A section) of their chosen topic. This will help develop and improve the student’s level of literacy as well as higher-order critical and creative thinking, abstraction and presentation abilities. INDIVIDUAL WRITING ASSIGNMENTS & IN-CLASS PRESENTATION: 25% Students must develop an extended piece of writing (~2500 words) regarding a chosen topic in intelligent wearable systems. Subsequently, students will also provide a short oral presentation overview of their chosen topic. This will help develop and improve the student’s level of literacy as well as higher-order critical and creating thinking, advanced literacy and comprehension and abstraction abilities.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Group Project Assignment	20%	✓	✓	✓	✓	✓	2. Individual Project Assignment	25%	✓	✓	✓	✓	✓	3. Pre-class preparation, attendance & participation	15%	✓	✓	✓	✓	✓	4. Exam	40%	✓	✓	✓	✓	✓	Total	100 %					
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3. Pre-class preparation, attendance & participation	15%	✓	✓	✓	✓	✓																																										
4. Exam	40%	✓	✓	✓	✓	✓																																										
Total	100 %																																															

	ATTENDANCE, PRE-CLASS PREPARATION & PARTICIPATION, PRESENTATIONS: 15% Pre-class activities (e.g.; short discussion topics questions preparation) [5% each]; class attendance [5%], &; active participation during the lecture presentation will be assessed (e.g.; via ‘clickers’ etc.) [5%]. EXAM PAPER: 40% A 3 hour, end-of-term exam paper comprising multiple long answer questions will test the breadth of knowledge and comprehension across the entirety of the course. The materials submitted for this assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students’ own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.		
Student Study Effort Expected	CLASS CONTACT:		
	▪ Lectures	18 hrs	
	▪ Workshops	19 hrs	
	OTHER STUDY EFFORT:		
	▪ Individual Assignment (<i>Due Week 6</i>)	24 hrs	
	▪ Group Assignment (<i>Due week 12</i>)	24 hrs	
	▪ Self-study & Class Preparation	20 hrs	
	TOTAL STUDENT STUDY EFFORT	105 hours	
Reading List and References	1. W. Barfield, Fundamentals of wearable computers and augmented reality, CRC Press, Taylor & Francis Group 2016. 2. X.M. Tao, Handbook of smart textiles, Springer 2015. 3. L. Shu, 2012. Wearable body area sensor networks for continuous dynamic health monitoring in daily activities: case study of intelligent footwear system, PhD thesis, Hong Kong Polytechnic University. 4. S. Cohen, Hal Rodriguez, Make it, wear it: wearable electronics for makers, crafters, and cosplayers, McGraw-Hill Education 2018. 5. J.D. Shi, S. Liu, L. Zhang, B. Yang, L. Shu, Y. Yang, M. Ren, Y. Wang, J.W. Chen, W. Chen, Y. Chai, X.M. Tao, 2019. Smart textile-integrated Microelectronic systems for wearable applications, Advanced Materials, DOI: 10.1002/adma.201901958, 1091958. 6. W. Zeng, L. Shu, Q. Li, S. Chen, F. Wang and X.M. Tao, 2014. Fiber-based wearable electronics: a review of materials, fabrication, devices and applications, Advanced Materials. 26(31):5310-5336, S. Schneegass, O. Amft, Smart Textiles - Fundamentals, Design, and Interaction, Springer, 2017. 7. S. Schneegass, O. Amft, Smart Textiles - Fundamentals, Design, and Interaction, Springer, 2017. 8. D.H. Kim, N.S. Lu ... J. Rogers, 2011. Epidermal electronics, Science, Vol. 333, Issue 6044, pp. 838-843.		

	9. J. McCann and D. Bryson, Smart clothes and wearable technology, Woodhead Pub., Institute of Textiles, CRC Press, 2009. 10. J. Holland. Wearable technology and mobile innovations for next-generation education, Information Science Reference 2016. 11. R. Tong, Wearable technology in medicine and health care, Elsevier/Academic Press 2018 12. https://my.clevelandclinic.org/health/treatments/17173-lifevest 13. https://teslasuit.io/
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Note 1: Intended Learning Outcomes

Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.